The function $f(x) = (x-3)^2 + \frac{1}{2}$ has domain $\mathrm{D}_f : (-\infty, \infty)$ and range $\mathrm{R}_f : \left[\frac{1}{2}, \infty\right)$

limits: $\lim\limits_{x \to a^-} f(x)$

derivatives: $f'(a) = \left.\dfrac{dy}{dx}\right|_{x=a} = \lim\limits_{x \to a} \dfrac{f(x) - f(a)}{x - a}$

integrals: $\displaystyle\int_{a^2}^{b^2} x^2\, dx = \left[\frac{x^3}{3}\right]_{a^2}^{b^2} = \frac{a^6}{3} - \frac{b^6}{3}$

sigma: $\displaystyle\sum_{n=1}^{\infty} ar^n = a + ar + ar^2 + \cdots + ar^n$

Riemann Integral: $\displaystyle\int_a^b f(x)\, dx = \lim_{x \to \infty} \sum_{k=1}^{n} f(x_k) \cdot \Delta x$

vector: $\vec{v} = v_1\vec{i} + v_2\vec{j} = \langle v_1, v_2 \rangle$